

E-Commerce Sales Analytics Dashboard Project Report

Abstract

This project presents a comprehensive **end-to-end E-Commerce Sales Analytics Dashboard** developed using SQL, Python, and Power BI. The primary objective is to transform raw transactional data into actionable business intelligence by analysing key aspects such as sales performance, product profitability, customer behaviour, regional trends, shipping patterns, and the critical impact of discounts on overall profitability.

The solution involves a complete data analytics pipeline including data collection, thorough data cleaning, Exploratory Data Analysis (EDA), advanced SQL querying for business metrics, statistical analysis in Python, and the creation of a dynamic, interactive Power BI dashboard. Various visualizations and analytical techniques were applied to uncover hidden patterns and trends within the Superstore e-commerce dataset.

Key findings from the analysis reveal that the **Technology category** drives the highest revenue, yet profitability is significantly influenced by discount strategies. The **Consumer segment** emerges as the most valuable customer group, while certain regions and states contribute disproportionately to total sales. The analysis also highlights a strong negative correlation between discount levels and profit margins, along with the identification of loss-making product sub-categories.

The final outcome of this project is a professionally designed, interactive Power BI dashboard featuring KPI cards, insightful charts, and dynamic slicers. This dashboard empowers business stakeholders to make data-driven decisions, optimize pricing and discount strategies, improve regional performance, and enhance overall profitability.

This project demonstrates the practical application of modern Business Intelligence tools in solving real-world e-commerce challenges and showcases proficiency in SQL, Python (Pandas, Matplotlib, Seaborn), and Power BI.

Table of Contents

Introduction	1 2.
Objectives	2 3.
Tools and Technologies Used	2 4.
Dataset Description	3 5.
Project Workflow	3 6.
Data Cleaning and Preparation	4 7.
SQL Analysis	4 8.
Python Exploratory Data Analysis	5 9.
Power BI Dashboard Development	6 10.
Business Insights	7 11.
Challenges Faced	8 12.
Learning Outcomes	8 13.
Conclusion	9 14.
Future Enhancements	9 15.
References	10

1. Introduction

In today's competitive digital marketplace, organizations generate vast volumes of transactional data daily. Effective analysis of this data is critical for understanding customer behaviour, optimizing profitability, identifying market trends, and enabling strategic decision-making.

This project performs in-depth sales analytics on e-commerce data using SQL for querying and aggregation, Python for exploratory analysis and visualization, and Power BI for interactive dashboard development.

2. Objectives

- Analyse overall e-commerce sales performance
- Identify profitable and loss-making product categories and sub-categories
- Evaluate customer segments and regional performance
- Examine the relationship between discounts and profits
- Develop meaningful visualizations and insights
- Build an interactive Power BI dashboard
- Strengthen practical skills in SQL, Python, and Power BI

3. Tools and Technologies Used

Tool/Technology	Purpose
SQL	Data querying and aggregation
Python	Data analysis and visualization
Pandas	Data manipulation
Matplotlib & Seaborn	Statistical visualization
Power BI	Interactive dashboard creation

4. Dataset Description

The dataset used in this project contains e-commerce sales transactions with details related to products, customers, sales, profits, discounts, and shipping information.

Dataset Columns

Column Name	Description
Ship Mode	Shipping method used
Segment	Customer segment
Country	Country of customer
City	Customer city
State	Customer state
Postal Code	Postal code
Region	Geographic region
Category	Product category
Sub-Category	Product sub-category
Sales	Revenue generated
Quantity	Number of items sold
Discount	Discount applied
Profit	Profit earned

5. Project Workflow

- Data Collection
- Data Cleaning and Preparation
- SQL Analysis
- Exploratory Data Analysis (Python)
- Data Visualization
- Power BI Dashboard Development
- Insight Generation

6. Data Cleaning and Preparation

Before analysis, the dataset was cleaned and prepared to ensure accurate insights.

- **Checking Missing Values**

Missing values were identified using Python functions to ensure data quality:

```
df.isnull().sum()
```

- **Removing Duplicate Records**

Duplicate rows were removed to avoid inaccurate aggregation and distorted insights.

```
df.drop_duplicates(inplace=True):
```

```
df.drop_duplicates(inplace=True)
```

- **Data Type Verification**

The dataset structure and column data types were verified using:

```
df.info():
```

```
df.info()
```

- **Statistical Summary**

Descriptive statistics were generated to understand data distribution.

```
df.describe():
```

```
df.describe()
```

This helped identify:

- sales variation
- profit distribution
- discount ranges
- outliers

7. SQL Analysis

- Total Sales and Total Profit
- Sales and Profit by Category
- Regional Sales Performance

- Discount vs Profit Analysis
- Loss-making Sub-Categories

8. Python Exploratory Data Analysis

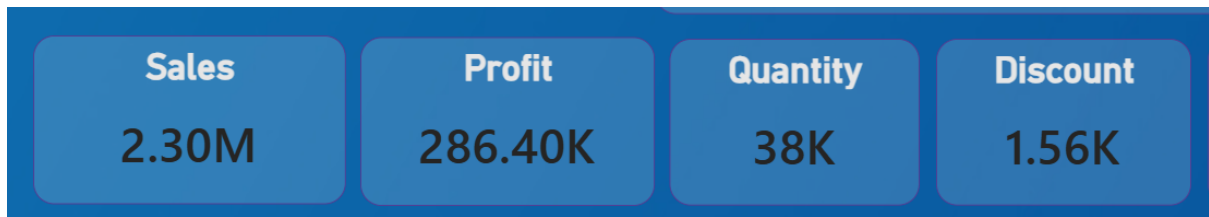
Key visualizations created using Matplotlib and Seaborn:

- Sales by Category (Technology highest)
- Profit by Category
- Region-wise Sales Distribution
- Top 10 States by Sales
- Discount vs Profit Scatter Plot (Negative Correlation)
- Segment-wise Sales (Consumer dominant)
- Shipping Mode Analysis

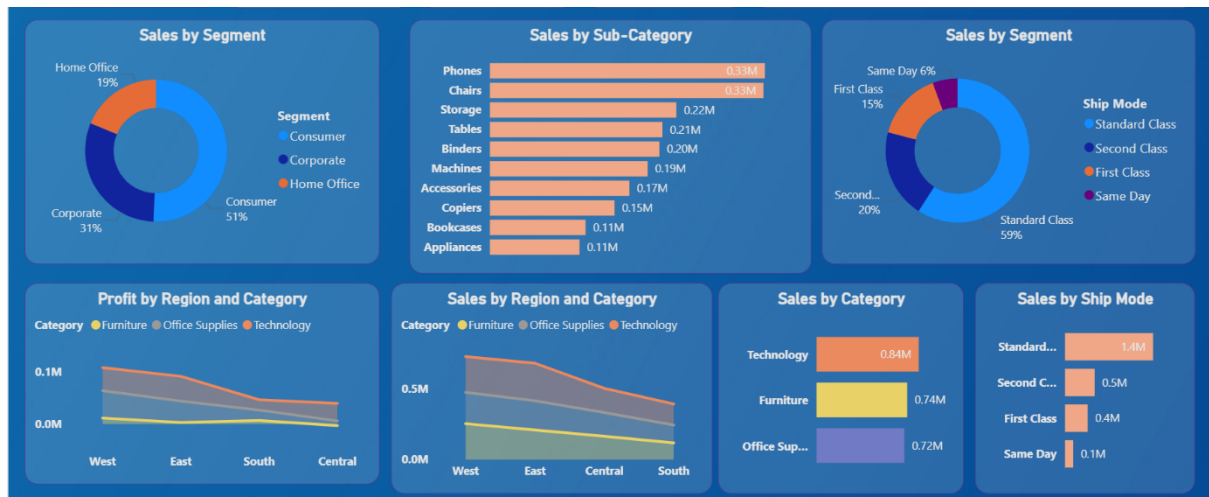
9. Power BI Dashboard Development



- KPI Cards: Sales, Profit, Quantity, Discount



- Bar Charts, Pie Charts



- Interactive Slicers (Region)



10. Business Insights

- Technology category generated the highest revenue.
- High sales do not always mean high profit.
- Higher discounts significantly reduce profitability.
- Consumer segment contributes the maximum sales.
- Sales are highly concentrated in a few states and regions.
- Standard Class is the most preferred shipping mode.
- Some sub-categories are making losses.

11. Challenges Faced

- Power BI relationship modeling and DAX measures

- Dashboard formatting and alignment
- Converting technical analysis into clear business insights

12. Learning Outcomes

- Advanced SQL querying and data aggregation
- Python data analysis using Pandas, Matplotlib & Seaborn
- Professional dashboard development in Power BI
- Business storytelling and insight generation

13. Conclusion

The E-Commerce Sales Analytics project successfully transformed raw sales data into actionable business intelligence using SQL, Python, and Power BI. The interactive dashboard serves as a powerful decision-making tool for monitoring performance and driving profitability.

14. Future Enhancements

- Sales Forecasting using Machine Learning
- Customer Lifetime Value Analysis
- Real-time Dashboard Integration
- Discount Optimization Recommendations

15. References

- Superstore Dataset
- Microsoft Power BI Documentation
- Pandas, Matplotlib & Seaborn Official Documentation

Project Completed By: NEHA NAGESH

Date: May 2026